

Neurobiological Foundations of Reading Acceleration: A Naturalistic fNIRS Brain Imaging Study

Lucas C. Gemal, Bruno Melo, Raimundo Soares Jr, Joao R. Sato, Roberto Lent

IDOR / UFRJ / UFABC, Rio de Janeiro, Brazil

Methods

Participants

Thirty-nine third- and fourth-grade primary school children were recruited from a single school in Rio de Janeiro, Brazil, during the first and second academic terms of 2025. Inclusion criteria required typical development, absence of auditory or visual impairments, no reported history of neurological or psychiatric disorder, and age-appropriate reading and writing performance as assessed by the School Achievement Test (TDE; Stein, 1994). Cognitive functioning was screened with the Wechsler Abbreviated Scale of Intelligence (WASI; Trentini et al., 2014); children with a full-scale IQ below 85 were excluded. Sociodemographic information was collected via a structured questionnaire administered to caregivers during the first week.

Following application of these criteria, 14 participants (mean age = 8.85 years, SD = 0.73; range 8-10 years) contributed complete datasets and were included in all analyses. Participants were randomly allocated to the Accelerated group (n = 7; 6 male, 1 female; mean age = 9.00 +/- 0.87 years) or the Non-Accelerated control group (n = 7; 4 male, 3 female; mean age = 8.70 +/- 0.59 years) using a random-number generator. The two groups did not differ significantly in age, IQ, or baseline reading measures. The included participants were identified as: SUBJ_001, SUBJ_002, SUBJ_004, SUBJ_005, SUBJ_006, SUBJ_007, SUBJ_018, SUBJ_024, SUBJ_027, SUBJ_028, SUBJ_030, SUBJ_034, SUBJ_035, and SUBJ_037. The study was approved by the institutional ethics committee, and written informed consent was obtained from all caregivers.

Experimental Protocol and Reading Acceleration Software

The longitudinal protocol comprised ten fNIRS-monitored sessions distributed across three weeks (two sessions per week), preceded by a psychometric assessment week. During the first week, the WASI, TDE, and the sociodemographic questionnaire were administered. Over the following three weeks, each participant completed one calibration session and nine training sessions (ac_01 through ac_09), all conducted in the school environment with the NIRSport2 portable fNIRS system recording continuously.

ACELETRA(R) is a gamified digital reading-acceleration tool designed for Brazilian Portuguese, implementing the principles of the Reading Acceleration Program (RAP; Breznitz et al., 2013). Sentences of 9-15 words were displayed on a touchscreen laptop and progressively masked from left to right using a visual pattern moving at individually calibrated speeds, obliging the reader to process text faster than their habitual pace. After each sentence, a comprehension question was presented with two response options. Stimulus presentation and response timing were recorded by the software for subsequent extraction of behavioural variables.

The calibration session comprised 4 blocks of 5 stimuli each (20 reading events per participant). Training sessions each comprised 7 blocks of 5 stimuli (35 reading events per

session). Blocks were separated by rest periods. Individual stimulus triggers for fNIRS analysis were constructed from event markers using a custom MATLAB script (contagem_nova.m), which identified block onset and offset from the first and last reading events within each rest-separated segment. Expected block counts per session were verified programmatically, with deviations flagged as warnings.

Behavioural Analyses

Behavioural performance was characterised by two session-level variables: median reading time (TimeRm, in milliseconds per character, $\text{ms} \cdot \text{char}^{-1}$) and reading accuracy (number of correct comprehension responses per session). TimeRm is inversely related to reading speed: lower values indicate faster, more proficient reading. To investigate longitudinal trajectories, separate linear mixed-effects models (LMMs) were fitted for TimeRm and accuracy using restricted maximum likelihood (REML) estimation. Each model included fixed effects for session number (continuous; 0-9, with 0 = calibration), group (Accelerated vs. Non-Accelerated; reference = Non-Accelerated), and their interaction, with a random intercept per participant. Models were fitted in Python 3.12 using statsmodels v. 0.14.6 (statsmodels.formula.api.mixedlm).

fNIRS Data Acquisition

Cortical haemodynamics were recorded continuously throughout all ACELETRA sessions using a NIRSport2 8-8 Core Unit (NIRx Medical Technologies, Berlin, Germany). The system simultaneously measured changes in oxyhaemoglobin (HbO), deoxyhaemoglobin (HbR), total haemoglobin (HbT), and oxygen saturation (StO₂) across 25 channels at a native sampling rate of 4.36 Hz, using source-detector pairs illuminated at 760 nm and 850 nm. The optode cap was positioned to cover left frontal and left temporal cortices, consistent with established reading neuroscience paradigms. HbO was the primary chromophore for all reported analyses; StO₂ results are reported as supplementary material.

Channels were grouped into two pre-defined regions of interest (ROIs). The frontal ROI comprised 7 channels (S1-D1, S2-D1, S2-D3, S1-D2, S3-D3, S4-D1, S3-D1) covering the left middle frontal gyrus and inferior frontal gyrus pars triangularis. The temporal ROI comprised 9 channels (S8-D7, S8-D4, S7-D7, S7-D4, S7-D6, S6-D4, S5-D5, S5-D6, S5-D4) covering left middle and inferior temporal gyri; channel S6-D5 was excluded from ROI-level analyses due to insufficient signal quality. All acquisition was conducted in the naturalistic school environment (child seated at a desk during software use), emphasising the ecological validity of the experimental design.

fNIRS Data Preprocessing

Preprocessing was performed in MATLAB using the NIRS Toolbox (nirs-toolbox.github.io), following a standardised eight-step pipeline applied identically to all 140 sessions (14 participants x 10 sessions). (1) Raw data were resampled to 4.00 Hz (nirs.modules.Resample; $F_s = 4$). (2) Baseline trimming was applied with a pre-stimulus

window of 0 s and a post-stimulus window of 10 s (nirs.modules.TrimBaseline). (3) Optical density was computed (nirs.modules.OpticalDensity). (4) Haemoglobin concentrations were derived using the modified Beer-Lambert Law (nirs.modules.BeerLambertLaw). (5) Total haemoglobin was calculated (nirs.modules.CalculateTotalHb), yielding four chromophore series per channel (HbO, HbR, HbT, StO₂; 100 channels total = 25 source-detector x 4 chromophores).

(6) A band-pass filter was applied (nirs.modules.BandPassFilter; high-pass = 0.01 Hz, low-pass = 0.10 Hz; do_downsample = 0), retaining the canonical haemodynamic frequency band. (7) Motion artefacts were corrected using a discrete wavelet transform filter (nirs.modules.WaveletFilter); stimulus markers were manually re-assigned to the filtered data post-hoc, as the WaveletFilter module does not propagate the stimulus structure. (8) First-level haemodynamic responses were estimated via a general linear model (nirs.modules.GLM; AddShortSepRegressors = 1), which included short-separation channel regressors to account for systemic physiological noise. The GLM yielded subject-level statistics (SubjStats) for each of the 140 sessions, providing per-channel, per-chromophore beta coefficients. Pipeline integrity was verified using a dedicated quality check script (relatorio_geral.m) that confirmed N = 140 sessions, Fs = 4.00 Hz, four chromophore types, 100 channels, and zero NaN or infinite values.

Neural Efficiency Index

Neural efficiency (NE) was operationalised following Curtin and Ayaz (2019), adapted for longitudinal fNIRS data. For each training session (sessions 1-9), the mean HbO beta coefficient across all channels within each ROI served as the neural activity estimate (beta). Both beta and reading time (TimeRm) were z-standardised prior to NE computation: z_beta was computed separately per ROI (frontal and temporal have distinct activation scales), whereas z_TimeRm was computed over the full training sample (all participants x sessions 1-9) to maintain a common performance reference. The NE index was then defined as:

$$NE = (-z_TimeRm - z_beta) / \sqrt{2}$$

The negative sign applied to z_TimeRm reflects the inverse relationship between reading time and performance (lower TimeRm = faster reading), such that higher NE values correspond to better performance relative to neural effort. Positive NE values indicate high performance with low activation; negative values indicate the inverse. The NE index was computed for all 14 participants across 9 training sessions and 2 ROIs, yielding 252 observations. Baseline values (calibration session) for beta and VELmed were retained as covariates (beta_baseline and VELmed_baseline) but excluded from z-standardisation.

To characterise longitudinal NE trajectories, separate LMMs were fitted for the frontal and temporal ROIs using REML estimation (statsmodels). Each model included fixed effects for group (reference = Non-Accelerated), session number (continuous, 1-9), their interaction, and baseline reading speed (VELmed at calibration) as a covariate. A random intercept per

participant was included. Each ROI model comprised $N = 126$ observations nested within 14 participants. Individual-level OLS regression slopes ($NE \sim sessao_num$) were additionally computed per participant x ROI to characterise heterogeneity, and group differences in slope distributions were evaluated via Fisher exact tests and independent samples t-tests (individual_ne_regressions.ipynb). Group differences in mean NE (collapsed across sessions) were assessed by Mann-Whitney U tests with rank-biserial correlation as the effect size measure.

Cortical Activation (beta) Analysis

Second-level group statistics were computed in MATLAB using the MixedEffects module (MixedEffects.m). The formula was: $\beta \sim -1 + group:experiment + (1|subject)$, with robust estimation enabled ($robust = 1$), yielding group-level beta estimates for each group x session cell without a common intercept.

To model longitudinal cortical activation trajectories and test group differences, a single Python LMM was fitted to per-ROI mean HbO beta values across all ten sessions ($N = 280$; $14 \times 10 \times 2$; ML estimation, $method = bfgs$). The model formula was: $\beta \sim group * sessao_num + roi + (1|subject)$, with ROI as an additive fixed covariate, a random intercept per participant, and the calibration session ($sessao_num = 0$) as the intercept anchor. The primary term of interest was the group x session interaction, representing the differential rate of beta change for the Accelerated group.

Frontal-Temporal Functional Connectivity

Inter-regional functional connectivity was computed in MATLAB using the Connectivity module of the NIRS Toolbox (conectividade_funcional.m). An autoregressive correlation function was applied ($nirs.sFC.ar_corr$; $window = 4 \times Fs = 16$ samples). Events were segmented per block ($divide_events = true$; minimum event duration = 10 s), and short-separation regressors were included ($AddShortSepRegressors = 1$). Pairwise Pearson correlations (R) and their Fisher-z transforms (Z) were computed for all channel pairs across 140 sessions.

Inter-regional connectivity was defined over pairs formed by one channel from the frontal ROI (6 channels: S1-D1, S2-D1, S2-D3, S1-D2, S3-D3, S3-D1) and one from the temporal ROI (10 channels: S8-D7, S8-D4, S7-D7, S7-D4, S7-D6, S6-D5, S6-D4, S5-D5, S5-D6, S5-D4), yielding 60 inter-regional pairs per session. The mean Fisher-z ($conexao_inter$) across all 60 pairs was the connectivity summary measure per participant x session. Three participant-session combinations were excluded due to insufficient signal quality (SUBJ_001, SUBJ_005, and SUBJ_024, all at session 4), yielding $N = 121$ observations. Baseline connectivity ($conn_baseline$, calibration session) was retained as a covariate. The LMM formula was: $conexao_inter \sim group * session + conn_baseline + (1|subject)$, estimated with REML ($statsmodels, method = bfgs$), using sessions 1-9 only.

Channel-Level Cortical Activation

To localise group differences at single-channel resolution, a separate LMM was fitted for each of 15 available HbO channels (S4-D1 excluded: insufficient observations). Each model used ML estimation (method = bfgs) with the formula: $\text{beta} \sim \text{group} * \text{sessao_num} + (1|\text{subject})$, including all 10 sessions (calibration as anchor). The key term was the group x sessao_num interaction. Multiple-comparison correction across 15 tests was applied using the Benjamini-Hochberg FDR procedure ($\alpha = 0.05$). Significant channels ($q < 0.05$) were mapped to approximate anatomical regions using the international 10-10 system (channel_lmm_anatomical.ipynb).

Correlation Between HbO Slope and Reading Time Slope

To examine neural-behavioural coupling at the individual level, Spearman rank correlations were computed between individual HbO beta slopes and reading time slopes for each of the six channels that survived FDR correction. For each participant x channel, OLS regression of HbO beta $\sim$ sessao_num was fitted over training sessions 1-9, extracting the slope (hbo_slope, in microM per session). An OLS regression of TimeRm $\sim$ sessao_num was fitted per participant to extract the reading time slope (timerm_slope, in ms*char⁻¹ per session). Spearman correlations were computed for three scopes: all participants (N = 13, excluding SUBJ_030 as a multivariate outlier per Cook's distance > 1), the Accelerated subgroup (n = 7), and the Non-Accelerated subgroup (n = 6). FDR correction (Benjamini-Hochberg) was applied separately within each scope across 6 channels.

General Statistical Considerations

All Python-based LMMs were estimated using statsmodels v. 0.14.6 (Python 3.12), with NumPy v. 2.4. REML estimation was used for models in which inference focused on fixed effects (neural efficiency, connectivity); ML estimation was used when likelihood-based model comparisons were performed or when nested models were compared (cortical activation, channel-level models). All models included only a random intercept per participant. The reference category for group was Non-Accelerated in all models. The significance threshold was $\alpha = 0.05$ for all tests. For channel-level analyses, multiple comparisons were controlled using the Benjamini-Hochberg FDR procedure. Outlier observations were identified using Cook's distance > 1 and removed prior to final model fitting where applicable. All reported confidence intervals are 95% Wald CIs.

Results

A total of 14 participants contributed valid data across all analyses. The complete dataset comprised 140 fNIRS sessions (14 participants x 10 sessions). Session-level exclusions are described for each analysis below.

Reading Speed and Accuracy

LMM of median reading time (TimeRm; lower values = faster reading) revealed a significant main effect of session ($\beta = -3.31 \text{ ms} \cdot \text{char}^{-1}$ per session, $p < 0.001$), reflecting a general reduction in reading time across training for both groups combined. The main effect of group was not significant. Critically, the session x group interaction was significant, indicating that the Accelerated group showed a steeper reduction in reading time across sessions relative to the Non-Accelerated group, confirming the training-induced acceleration in reading fluency.

Analysis of reading accuracy revealed a significant session x group interaction ($\beta = -0.014$, $SE = 0.003$, $z = -4.71$, $p < 0.001$, 95% CI $[-0.020, -0.008]$), indicating that the Accelerated group improved comprehension accuracy across sessions, while the Non-Accelerated group showed no significant change ($\beta = 0.003$, $SE = 0.002$, $z = 1.26$, $p = 0.209$).

Neural Efficiency

For the frontal ROI ($N = 126$, 14 participants), the LMM intercept was significant ($\beta = 1.131$, $SE = 0.500$, $z = 2.263$, $p = 0.024$, 95% CI $[0.151, 2.110]$). Baseline reading speed (VELmed_baseline) was significantly and negatively associated with neural efficiency ($\beta = -0.012$, $SE = 0.003$, $z = -4.053$, $p < 0.001$, 95% CI $[-0.018, -0.006]$), indicating that faster baseline readers showed lower frontal NE, consistent with ceiling effects in more fluent children. The session main effect ($\beta = 0.046$, $SE = 0.035$, $z = 1.297$, $p = 0.194$) and the group x session interaction ($\beta = 0.067$, $SE = 0.050$, $z = 1.352$, $p = 0.176$, 95% CI $[-0.030, 0.164]$) were not significant for the frontal ROI.

For the temporal ROI ($N = 126$, 14 participants), the session x group interaction was statistically significant ($\beta = 0.109$, $SE = 0.047$, $z = 2.298$, $p = 0.022$, 95% CI $[0.016, 0.202]$), indicating that the Accelerated group exhibited a progressive increase in temporal neural efficiency across training relative to controls. Baseline reading speed remained a significant negative predictor (VELmed_baseline: $\beta = -0.010$, $SE = 0.003$, $z = -2.907$, $p = 0.004$, 95% CI $[-0.017, -0.003]$). The group main effect ($\beta = -0.044$, $SE = 0.371$, $z = -0.119$, $p = 0.906$) and session main effect ($\beta = 0.027$, $SE = 0.034$, $z = 0.810$, $p = 0.418$) were not significant. Full LMM results are presented in Table 2.

Individual OLS slopes ($NE \sim \text{session}$) confirmed the temporal ROI pattern: Accelerated group showed a higher mean slope ($M = 0.136$; 3 of 7 participants significant at $p < 0.05$) compared to Non-Accelerated ($M = 0.027$; 2 of 7 significant), with a trend approaching significance (t-test: $p = 0.086$). In the frontal ROI, slopes were positive in both groups

(Accelerated: $M = 0.113$; Non-Accelerated: $M = 0.046$) but did not differ significantly (t-test: $p = 0.198$). Individual slope estimates are presented in Table 5.

Mann-Whitney U tests on mean NE (sessions 1-9, per participant) revealed large and significant group differences in both ROIs. Frontal ROI: Accelerated $M = 0.470$ ($SD = 0.286$) vs. Non-Accelerated $M = -0.470$ ($SD = 0.762$); $U = 46$, $p = 0.004$, $r = -0.878$ (large effect). Temporal ROI: Accelerated $M = 0.543$ ($SD = 0.482$) vs. Non-Accelerated $M = -0.543$ ($SD = 0.725$); $U = 47$, $p = 0.002$, $r = -0.918$ (large effect).

*Table 2. Linear mixed-effects model results for neural efficiency by ROI. REML estimation; reference group = Non-Accelerated. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. $N = 126$ per ROI.*

Parameter	Frontal beta	Frontal SE	Frontal p	Temporal beta	Temporal SE	Temporal p	Temporal 95% CI
Intercept	1.131*	0.500	0.024	0.906	0.562	0.107	[-0.195, 2.007]
Group [Accel.]	0.091	0.352	0.796	-0.044	0.371	0.906	[-0.771, 0.683]
Session (sessao_num)	0.046	0.035	0.194	0.027	0.034	0.418	[-0.039, 0.093]
Group x Session	0.067	0.050	0.176	0.109*	0.047	0.022	[0.016, 0.202]
VELmed baseline	-0.012***	0.003	<0.001	-0.010**	0.003	0.004	[-0.017, -0.003]

Cortical Activation (beta)

The LMM of HbO beta across all sessions and ROIs ($N = 280$; ML estimation) yielded no statistically significant fixed effects. The intercept, representing the Non-Accelerated group's mean beta in the frontal ROI at calibration, was not significant (beta = -0.318 , $SE = 0.517$, $z = -0.615$, $p = 0.539$, 95% CI $[-1.331, 0.695]$). The group main effect (beta = -0.126 , $SE = 0.693$, $z = -0.181$, $p = 0.856$) and the ROI main effect (beta = 0.463 , $SE = 0.329$, $z = 1.408$, $p = 0.159$) were not significant. The session progression term was not significant (beta = 0.036 , $SE = 0.081$, $z = 0.443$, $p = 0.658$). The group x session interaction (training-induced beta change) was negative but non-significant (beta = -0.118 , $SE = 0.115$, $z = -1.034$, $p = 0.301$, 95% CI $[-0.343, 0.106]$).

Frontal-Temporal Functional Connectivity

The inter-regional connectivity dataset comprised 121 observations after excluding three sessions with poor signal quality (SUBJ_001, SUBJ_005, and SUBJ_024, all at session 4). Descriptive statistics indicated comparable mean connectivity across the training period between groups (Accelerated: $M = 0.347$, $SD = 0.186$; Non-Accelerated: $M = 0.339$, $SD = 0.176$, expressed as mean Fisher-z, corresponding to r_{mean} approximately 0.33-0.35).

The LMM (REML; $N = 121$, 14 participants) showed no significant group x session interaction (beta = 0.004 , $SE = 0.012$, $z = 0.315$, $p = 0.753$, 95% CI $[-0.020, 0.028]$). The group main effect was not significant (beta = -0.050 , $SE = 0.091$, $z = -0.552$, $p = 0.581$). The session main effect showed a marginally non-significant positive trend (beta = 0.015 , $SE =$

0.009, $z = 1.777$, $p = 0.075$, 95% CI [-0.002, 0.032]), suggesting a weak tendency for connectivity to increase across sessions in both groups. Baseline connectivity (conn_baseline) was not a significant predictor ($\beta = 0.284$, $SE = 0.285$, $z = 0.999$, $p = 0.318$). Full LMM results are presented in Table 3.

Table 3. Linear mixed-effects model results for frontal-temporal functional connectivity (Fisher-z transformed). REML estimation; reference group = Non-Accelerated; $N = 121$ observations.

Parameter	beta	SE	z	p	95% CI
Intercept	0.212	0.113	1.872	0.061	[-0.010, 0.434]
Group [Accel.]	-0.050	0.091	-0.552	0.581	[-0.229, 0.129]
Session	0.015	0.009	1.777	0.075	[-0.002, 0.032]
Group x Session	0.004	0.012	0.315	0.753	[-0.020, 0.028]
conn_baseline	0.284	0.285	0.999	0.318	[-0.274, 0.842]

Channel-Level Cortical Activation

Channel-level LMMs revealed 6 channels with significant group x session interactions surviving FDR correction ($q < 0.05$ out of 15 tests), comprising 1 frontal and 5 temporal channels (Table 4). All significant channels except S5-D5 showed negative interaction coefficients, indicating a steeper decline in HbO beta for the Accelerated group across training sessions.

The significant frontal channel, S3-D1 (10-10: AF3-F5; left IFG pars triangularis, ~22%), showed $\beta = -0.296$, $p = 0.003$, $q = 0.014$, 95% CI [-0.490, -0.103]. Of the five temporal channels, four showed negative interactions: S5-D4 (left middle temporal gyrus, ~41%; $\beta = -0.297$, $p = 0.002$, $q = 0.014$, 95% CI [-0.481, -0.112]), S7-D7 (left middle temporal gyrus, ~55%; $\beta = -0.298$, $p = 0.002$, $q = 0.014$, 95% CI [-0.490, -0.106]), S5-D6 (left middle temporal gyrus, ~47%; $\beta = -0.318$, $p = 0.006$, $q = 0.021$, 95% CI [-0.547, -0.090]), and S7-D4 (left middle temporal gyrus, ~72%; $\beta = -0.197$, $p = 0.009$, $q = 0.023$, 95% CI [-0.346, -0.049]). Channel S5-D5 (left inferior temporal gyrus, ~29%) showed a positive interaction ($\beta = +0.215$, $p = 0.007$, $q = 0.021$, 95% CI [0.059, 0.372]), indicating greater HbO increase in the Accelerated group; this channel was treated separately in the neural-behavioural correlation analysis.

Table 4. Channels with significant group x session interactions surviving FDR correction (channel-level LMM, HbO). β = interaction coefficient (Accelerated minus Non-Accelerated trajectory); q = Benjamini-Hochberg-corrected p-value; 15 channels tested.

Channel	10-10	Region	ROI	beta	p	q	95% CI
S3-D1	AF3-F5	L IFG p.Triangularis (22%)	FRONTAL	-0.296	0.003	0.014	[-0.490, -0.103]
S5-D4	P7-TP7	L Middle Temporal Gyrus (41%)	TEMPORAL	-0.297	0.002	0.014	[-0.481, -0.112]
S7-D7	T7-C5	L Middle Temporal Gyrus (55%)	TEMPORAL	-0.298	0.002	0.014	[-0.490, -0.106]

S5-D6	P7-P5	L Middle Temporal Gyrus (47%)	TEMPORAL	-0.318	0.006	0.021	[-0.547, -0.090]
S5-D5	P7-P9	L Inferior Temporal Gyrus (29%)	TEMPORAL	+0.215	0.007	0.021	[0.059, 0.372]
S7-D4	CP5-TP7	L Middle Temporal Gyrus (72%)	TEMPORAL	-0.197	0.009	0.023	[-0.346, -0.049]

Correlation Between HbO Slope and Reading Time Slope

Spearman rank correlations between individual HbO beta slopes and TimeRm slopes were computed for the 6 FDR-significant channels. Across the combined sample (N = 13, SUBJ_030 excluded as outlier), all correlations were negative and non-significant after FDR correction (all $q > 0.60$). The strongest nominal association was at S7-D4 ($\rho = -0.456$, $p = 0.117$, $q = 0.629$), followed by S5-D5 ($\rho = -0.319$, $p = 0.289$), S5-D6 ($\rho = -0.236$, $p = 0.437$), S3-D1 ($\rho = -0.159$, $p = 0.603$), S7-D7 ($\rho = -0.154$, $p = 0.616$), and S5-D4 ($\rho = -0.148$, $p = 0.629$). The direction of all correlations (with the exception of S5-D5 when analysed in the full group) was consistent with a neural efficiency pattern: subjects with steeper HbO decline tended to show greater reading time reduction, though the associations did not reach statistical significance.

Within the Non-Accelerated subgroup ($n = 6$), a nominally significant negative correlation was observed at S7-D4 ($\rho = -0.886$, $p = 0.019$), which did not survive FDR correction ($q = 0.113$). Within the Accelerated subgroup ($n = 7$), no significant correlations were found (all $p > 0.28$).

Table 5. Individual OLS regression slopes for neural efficiency (NE ~ session, training sessions 1-9) per participant and ROI. Positive slopes = increasing NE across training. p-values from individual OLS models.

Subject	Group	ROI	Slope	p-value	R2	Sig.
SUBJ_002	Accelerated	FRONTAL	0.193	0.016	0.590	$p < 0.05$
SUBJ_002	Accelerated	TEMPORAL	0.106	0.085	0.364	$p < 0.10$
SUBJ_004	Accelerated	FRONTAL	0.196	0.229	0.199	n.s.
SUBJ_004	Accelerated	TEMPORAL	0.257	0.001	0.811	$p < 0.05$
SUBJ_006	Accelerated	FRONTAL	-0.012	0.873	0.004	n.s.
SUBJ_006	Accelerated	TEMPORAL	-0.003	0.979	0.000	n.s.
SUBJ_007	Accelerated	FRONTAL	0.107	0.656	0.030	n.s.
SUBJ_007	Accelerated	TEMPORAL	0.232	0.034	0.497	$p < 0.05$
SUBJ_027	Accelerated	FRONTAL	0.053	0.389	0.108	n.s.
SUBJ_027	Accelerated	TEMPORAL	0.198	0.002	0.755	$p < 0.05$
SUBJ_028	Accelerated	FRONTAL	0.110	0.061	0.415	$p < 0.10$
SUBJ_028	Accelerated	TEMPORAL	0.173	0.108	0.327	n.s.
SUBJ_034	Accelerated	FRONTAL	0.142	0.053	0.437	$p < 0.10$
SUBJ_034	Accelerated	TEMPORAL	-0.010	0.949	0.001	n.s.
SUBJ_001	Non-Accel.	FRONTAL	0.068	0.354	0.123	n.s.
SUBJ_001	Non-Accel.	TEMPORAL	0.005	0.920	0.002	n.s.
SUBJ_005	Non-Accel.	FRONTAL	0.003	0.945	0.001	n.s.
SUBJ_005	Non-Accel.	TEMPORAL	0.025	0.738	0.017	n.s.
SUBJ_018	Non-Accel.	FRONTAL	0.078	0.138	0.286	n.s.

SUBJ_018	Non-Accel.	TEMPORAL	0.001	0.993	0.000	n.s.
SUBJ_024	Non-Accel.	FRONTAL	0.115	0.204	0.219	n.s.
SUBJ_024	Non-Accel.	TEMPORAL	0.169	0.001	0.801	p < 0.05
SUBJ_030	Non-Accel.	FRONTAL	-0.116	0.124	0.303	n.s.
SUBJ_030	Non-Accel.	TEMPORAL	-0.130	0.219	0.207	n.s.
SUBJ_035	Non-Accel.	FRONTAL	-0.040	0.515	0.063	n.s.
SUBJ_035	Non-Accel.	TEMPORAL	-0.049	0.664	0.029	n.s.
SUBJ_037	Non-Accel.	FRONTAL	0.210	0.017	0.582	p < 0.05
SUBJ_037	Non-Accel.	TEMPORAL	0.169	0.023	0.546	p < 0.05